

PCAccess — Full Project Presentation

Enterprise Remote PC Access, Bi-Directional Streaming & Virtual Drive Ecosystem

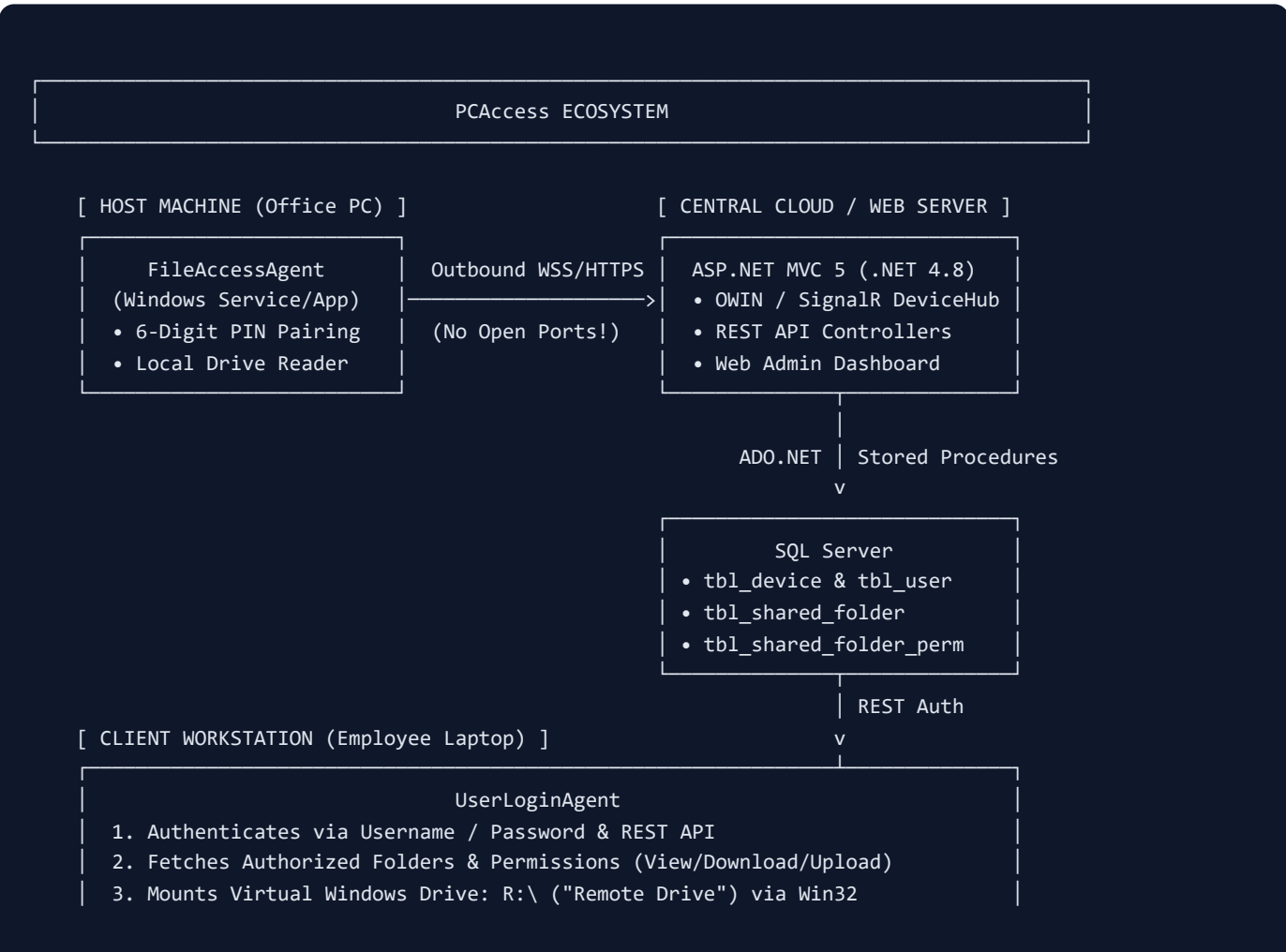
- ARCHITECTURE & WORKFLOW
- TECH STACK: .NET 4.8 + SIGNALR
- ZERO-TRUST OUTBOUND
- DUAL-AGENT SYSTEM

1. Executive Summary: What is PCAccess?

PCAccess ek secure, enterprise-ready remote file sharing aur PC access platform hai jo businesses ko without VPN, without complex port-forwarding aur without public IP ke unke remote machines ke physical folders ko centrally connect aur manage karne ki suvidha deta hai.

Key Innovation: Central web dashboard se devices aur folder permissions manage hoti hain, jabki desktop par **UserLoginAgent** automatically local virtual drive **R: ("Remote Drive")** mount kar deta hai, jisse employees ko lagta hai ki files unke apne PC me local folder ki tarah open ho rahi hain.

2. End-to-End System Architecture



4. User browses shared files natively in Windows Explorer!

3. Complete Technology Stack

Component	Technology Used	Role & Technical Rationale
Web Application	ASP.NET MVC 5 (.NET Framework 4.8)	Server-side rendering, robust routing, session management, secure controllers (AdminController, DeviceApiController, UserAgentApiController).
Real-Time Messaging	Microsoft SignalR 2.4.3 (OWIN)	Persistent full-duplex WebSocket connection between web browser and host agents. Handles device heartbeats, online/offline status, and streaming.
Frontend UI	Razor Views + Pure CSS (Design System) + Vanilla JS	Corporate responsive UI without heavy CSS libraries; SignalR client (jquery.signalR-2.4.3.min.js) for real-time live pulses.
Host Desktop Agent	C# Console / Windows Service (FileAccessAgent)	Runs in background on the PC hosting files. Connects outbound to SignalR Hub; generates temporary 6-digit pairing code.
Client Desktop Agent	C# Console App (UserLoginAgent)	Runs on employee desktop. Authenticates via REST, creates staging junctions, and mounts virtual drive R: in Windows Explorer.
Drive Virtualization	Win32 APIs (DefineDosDevice, subst, SHChangeNotify)	Native Windows OS kernel-level drive mapping without needing expensive third-party driver installations.
Database Layer	Microsoft SQL Server + Pure Stored Procedures	Zero raw SQL in C# code. All actions encapsulated in procedures (proc_device_manager, proc_user_agent_auth, etc.) with strict manual execution policy.
Serialization	Newtonsoft.Json (v13.0.3)	High-speed JSON serialization for network payloads and local configuration files (agent_config.json, user_agent_config.json).

4. Step-by-Step Business Workflow

- Host Registration & Pairing:** Host PC par FileAccessAgent start hota hai aur ek temporary 6-digit PIN screen par aata hai. Admin web portal me ye PIN enter karke device pair karta hai. Device permanently bind ho jata hai.
- Folder Sharing & Permission Masking:** Admin kisi bhi connected host PC ke folders ko register karta hai aur users/roles ke liye granular rights set karta hai (View, Download, Upload).
- Employee 1-Click Desktop Login:** Employee apne PC par UserLoginAgent launch karta hai aur apne credentials enter karta hai. System uske role aur permissions ko verify karta hai.
- Native Drive R: Mount:** User ke PC me instant virtual drive R: mount hoti hai. Employee kisi bhi standard Windows software (Excel, Word, Explorer) se in folders ko directly access kar sakta hai.
- Session Control & Logout:** Jab user ka kaam khatam ho, to console me L ya Q press karke ya logout.bat chala kar drive unmount aur safe logout ho jata hai.

5. Security & Zero-Trust Architecture

Outbound-Only Communication

Host PC aur Client PC dono sirf outbound HTTPS/WSS (Port 443) traffic bhejte hain. Router me koi bhi incoming port open karne ki zaroorat nahi hoti.

Granular Permission Enforcement

Database level par Stored Procedures har folder ke liye View, Download aur Upload rights check karte hain. Users sirf authorized data hi access kar sakte hain.